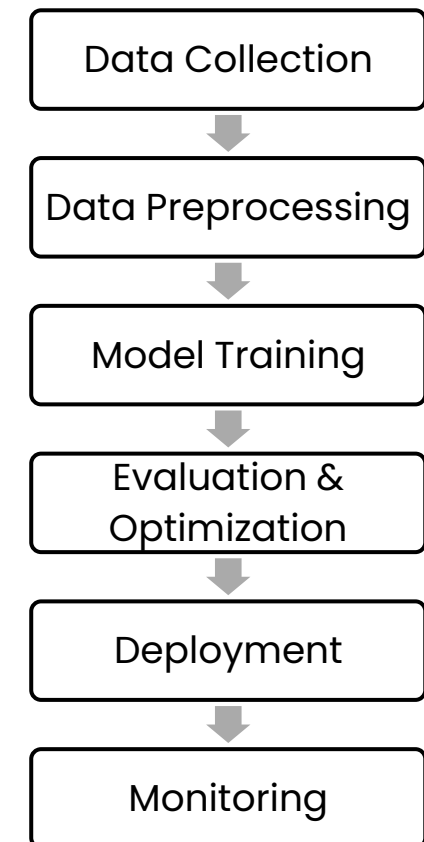
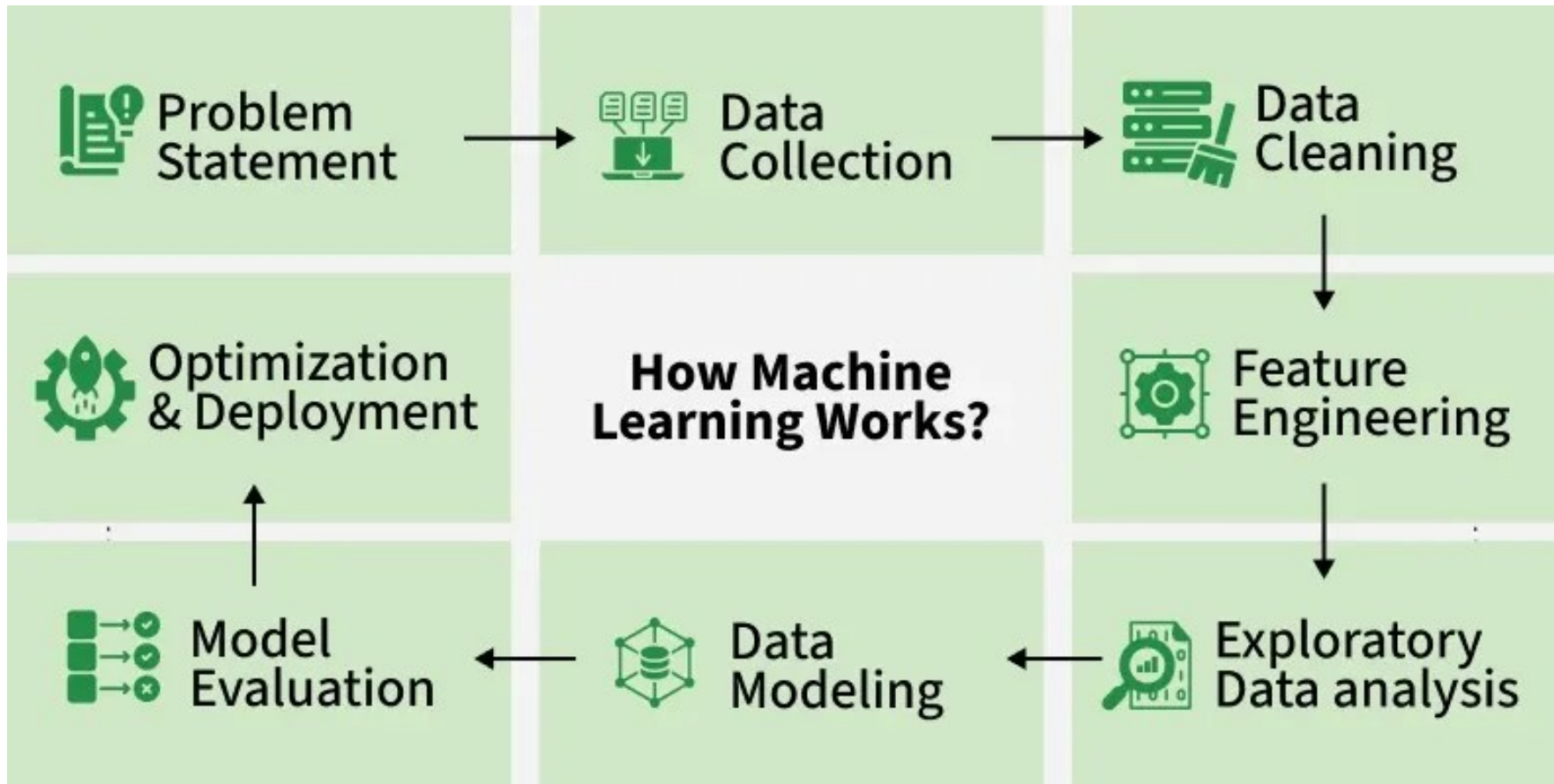


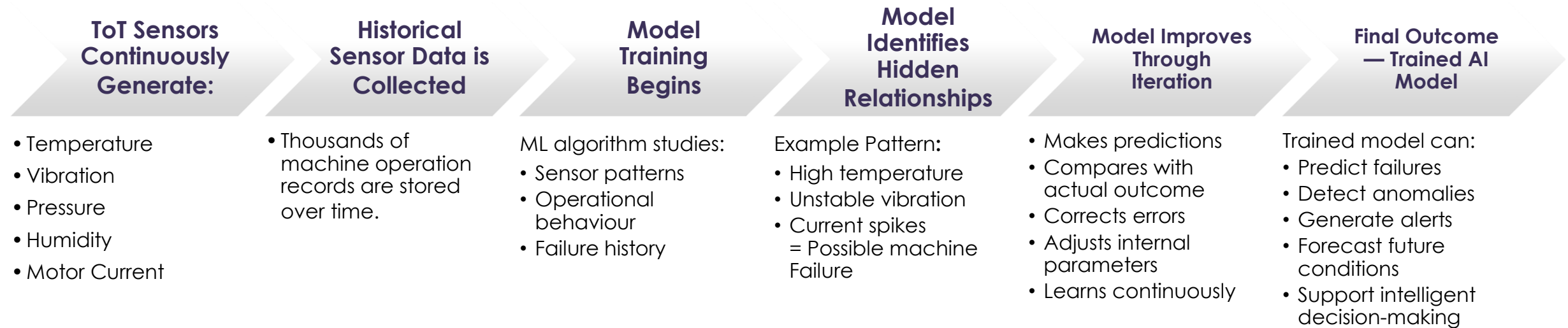
Machine Learning Process for AIOT

Machine learning Model learns patterns from data to make predictions without explicit programming. It improves accuracy over time by minimizing errors through training.



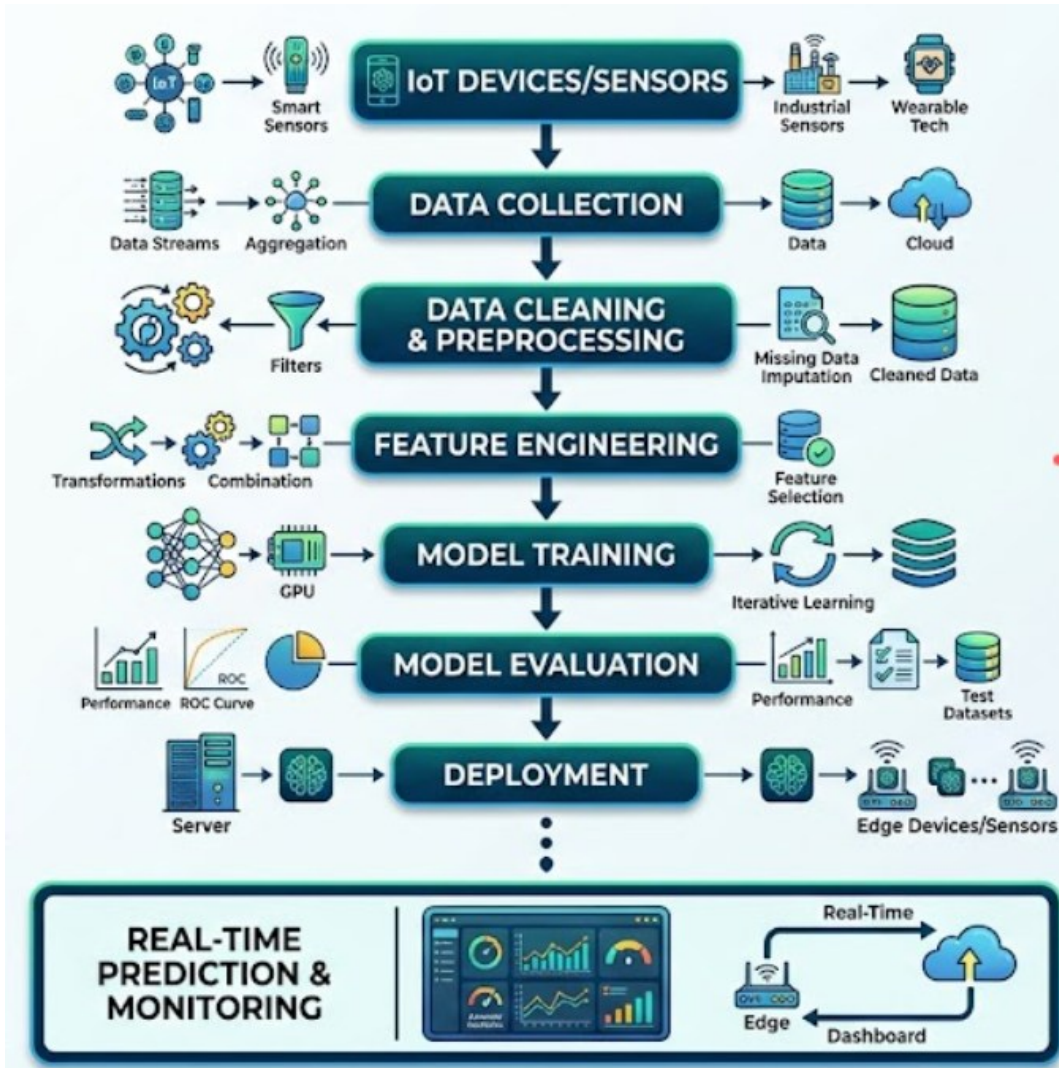
How an ML Model Learns – IoT Example

Scenario :- Multiple sensors attached to an industrial machine continuously generating data such as temperature, vibration, humidity, pressure, and motor current. Over time, the system collects thousands of historical records from the machine including machine operational data and failures after certain conditions.



Historical Data + Training + Error Correction = Intelligent Prediction Model

Data Collection, Data Cleaning & Preprocessing



Data Collection - Data is the Fuel for AI

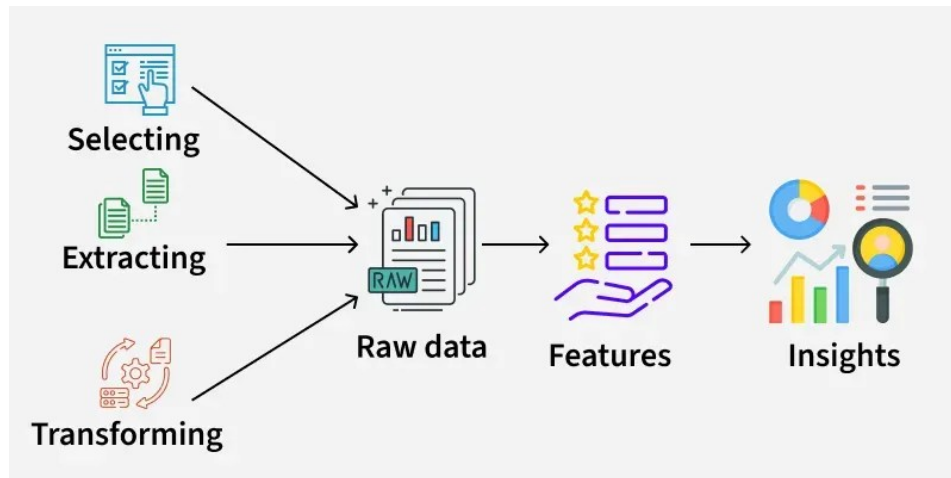
- Temperature sensors
- Vibration sensors
- Cameras
- Smart meters
- Industrial devices

Cleaning & Preprocessing - Preparing Data for Machine Learning

- Handling missing values
- Removing anomalies
- Noise filtering
- Normalization
- Formatting data

The Concept of Feature Engineering

Feature engineering transforms raw data into meaningful input variables (features) that help ML models learn patterns more effectively.



Raw Data :- temperature, vibration, pressure, motor current, and timestamps.



Feature Engineering

- temperature change over time,
- average temperature over the last 10 minutes
- maximum vibration during the last hour
- rate of pressure increase
- difference between current and previous readings

Model predicting	Features
Risk of cardiac disease	Age, Gender, Weight, Smoking habit, diabetic, etc.
If person is suitable for job	Educational qualification, years of experience
size of a shirt for a person	Age, gender, height, weight, etc.

The Concept of Feature Engineering

Timestamp (raw data)

	generate_series timestamp with time zone
1	2020-03-23 13:30:00+05:30
2	2020-03-23 18:30:00+05:30
3	2020-03-23 23:30:00+05:30
4	2020-03-24 04:30:00+05:30
5	2020-03-24 09:30:00+05:30

Features – Time indicators

hour of day

day of week

working shift

weekend

Analytical Inference by model based on feature Engineering

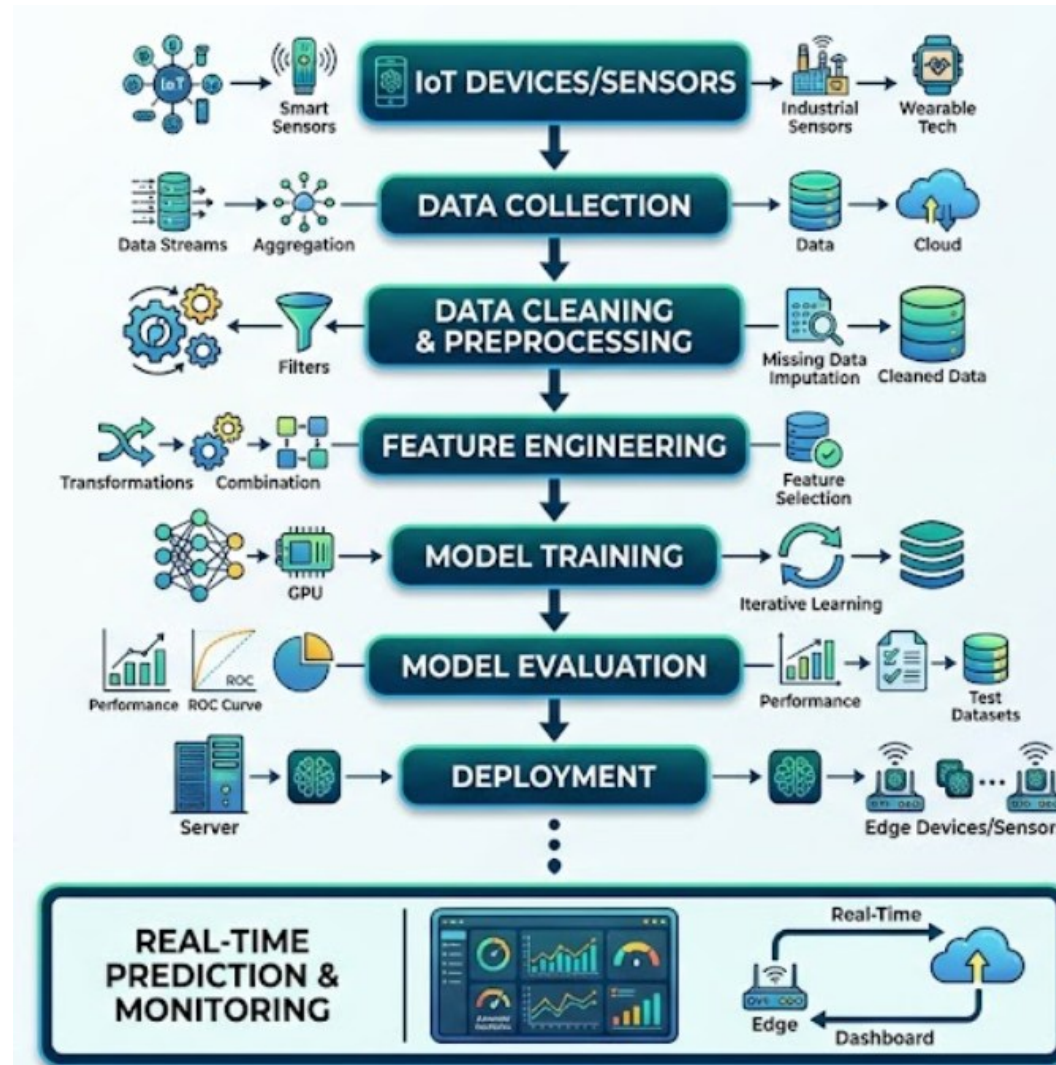
- certain machines consume more energy during night shifts
- failures occur more frequently during peak operational hours.

Feature engineering is the bridge between raw IoT data and intelligent AI predictions

- Feature Engineering converts raw sensor data into meaningful business intelligence.
- Many times, success of an AI project depends more on good feature engineering than on using complex ML algorithms.
- A simple algorithm with excellent features often performs better than an advanced algorithm with poor-quality data.

ML Workflow in AIOT

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Model Training

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Historical Data + Algorithm + Learning = Trained Model

What Happens During Model Training?

Machine Learning model studies historical IoT sensor data to learn:

- Patterns, Relationships, Trends, Operational behavior,
- Objective is to enable the model to make intelligent predictions on new live sensor data.

Historical Sensor Inputs

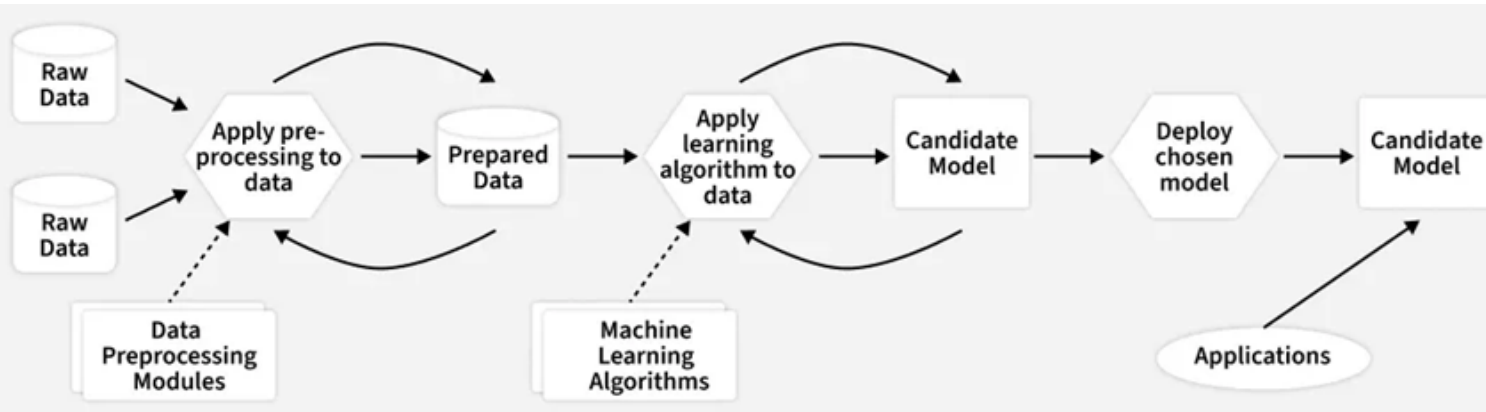
- Temp., Vibration, Motor Current, Machine Status

Model Training

- Feed historical sensor data to the algorithm
- Model makes predictions
- Compare prediction with actual outcome
- Adjust internal parameters to reduce errors
- Repeat thousands of times until learning improves

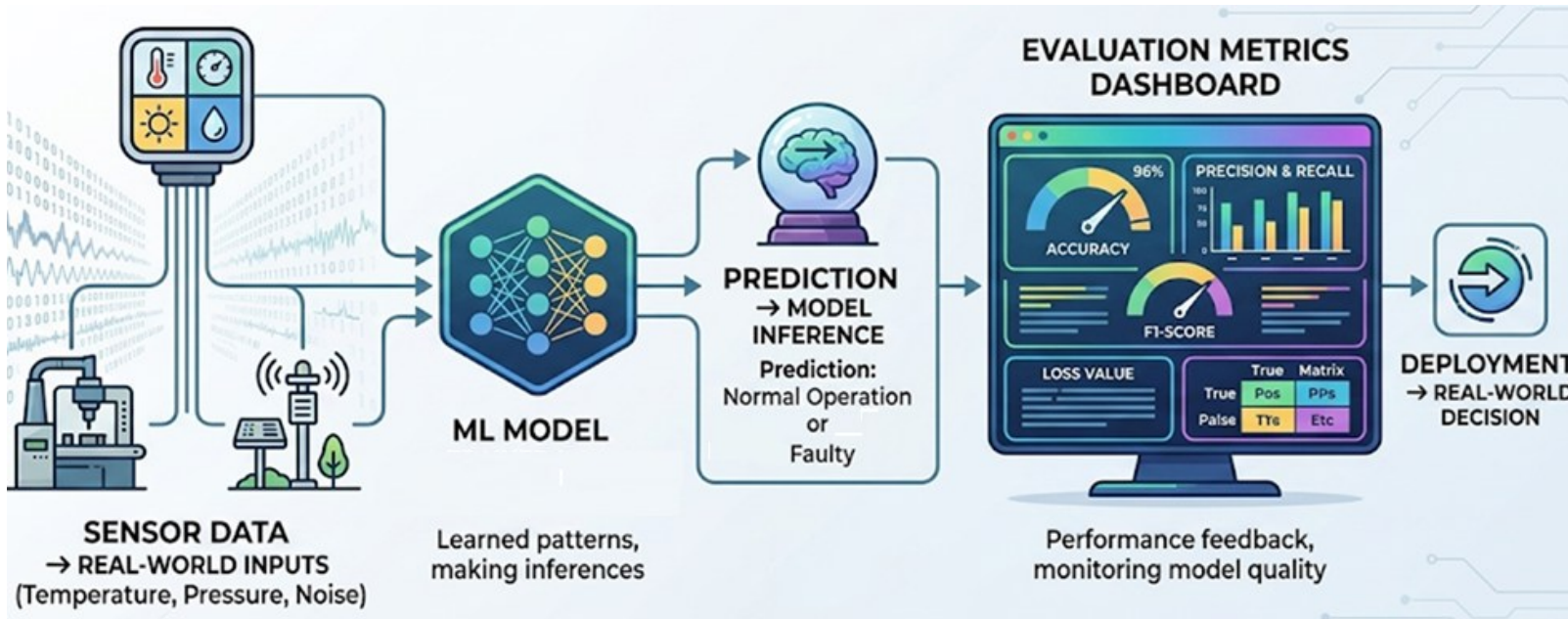
Output = Trained Model

- Deploy trained model on live System
- Model can now - Predict failures, Detect anomalies, Forecast future conditions



Model Evaluation

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Training Alone is Not Enough.. We must measure

- How accurate the model is
- How often it makes mistakes
- Whether predictions are reliable
- If the model can be trusted in production

IIoT Example — Predictive Maintenance

Model Objective - Predict whether an industrial machine is:

- Healthy
- Warning
- Failure Risk

Input Data

- Temperature
- Vibration
- Pressure
- Motor Current

Model Prediction
“Possible Machine Failure”

Evaluation Question
Was the prediction actually correct ?

Model Evaluation

A Model is Valuable only if its predictions are reliable in real-world AIoT systems.

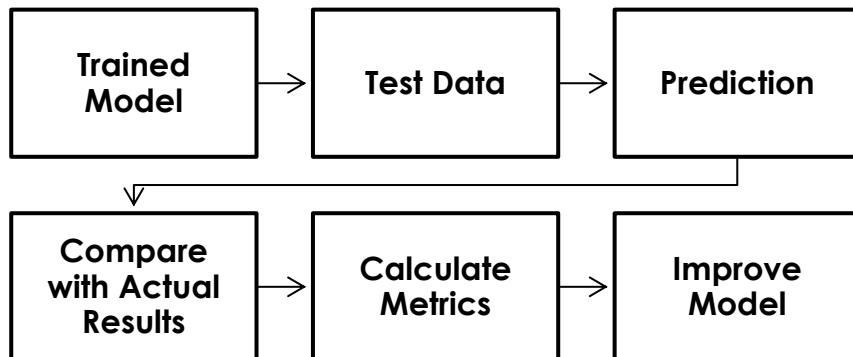
Goal of Model Evaluation

- Build a Reliable & Trustworthy AI System

Good AI Model Should:

- Detect real problems accurately
- Minimize false alerts
- Work reliably in real-world environments
- Perform consistently on live IoT data

Evaluation Lifecycle



Real AIoT Challenge

False Alarm Example

- Model predicts machine failure
- But machine is healthy

Impact

- Unnecessary maintenance
- Operational downtime
- Increased cost

Missed Failure Example

- Model predicts machine is healthy
- But machine fails

Impact

- Equipment damage
- Production loss
- Safety risk

Metric	Meaning	AIoT Importance
Accuracy	Overall correct predictions	General performance
Precision	Correct failure alerts	Reduces false alarms
Recall	Detecting actual failures	Prevents missed failures
F1 Score	Balance of Precision & Recall	Reliable prediction quality
MAE	Average prediction error	Useful in forecasting
R ² Score	Learning effectiveness	Measures pattern understanding

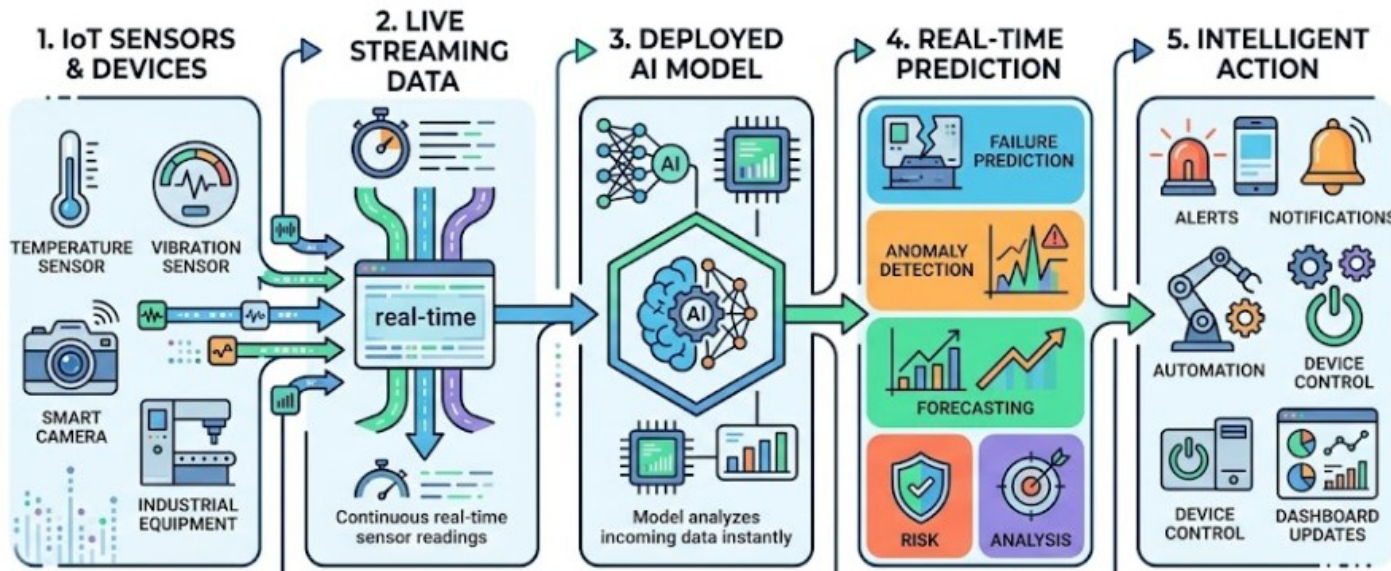
Deployment & Real-Time AI in AIOT

10

What Happens After Model Training?

- The trained AI model is deployed into real-world AIoT systems to make live predictions on streaming sensor data.

Real-Time AIoT Intelligence Flow



AIoT Example — Smart Factory

Real-Time Predictive Maintenance

Live Machine Sensor Data:

- Temperature rising, Vibration unstable
- Current increasing



AI Model Detects:

- “High probability of machine failure”



System Action:

- Generate alert , Notify operator, Schedule maintenance, Prevent breakdown

Deployment & Real-Time AI in AIOT

AIoT Systems
Must Be

Fast

Continuous

Adaptive

Scalable

Reliable

Deployment Environments

Deployment Type

AIoT Usage

Cloud AI

Centralized analytics

Edge AI

Low-latency local prediction

Hybrid AI

Edge + Cloud intelligence



Deployment transforms trained AI models into live intelligent AIoT systems.

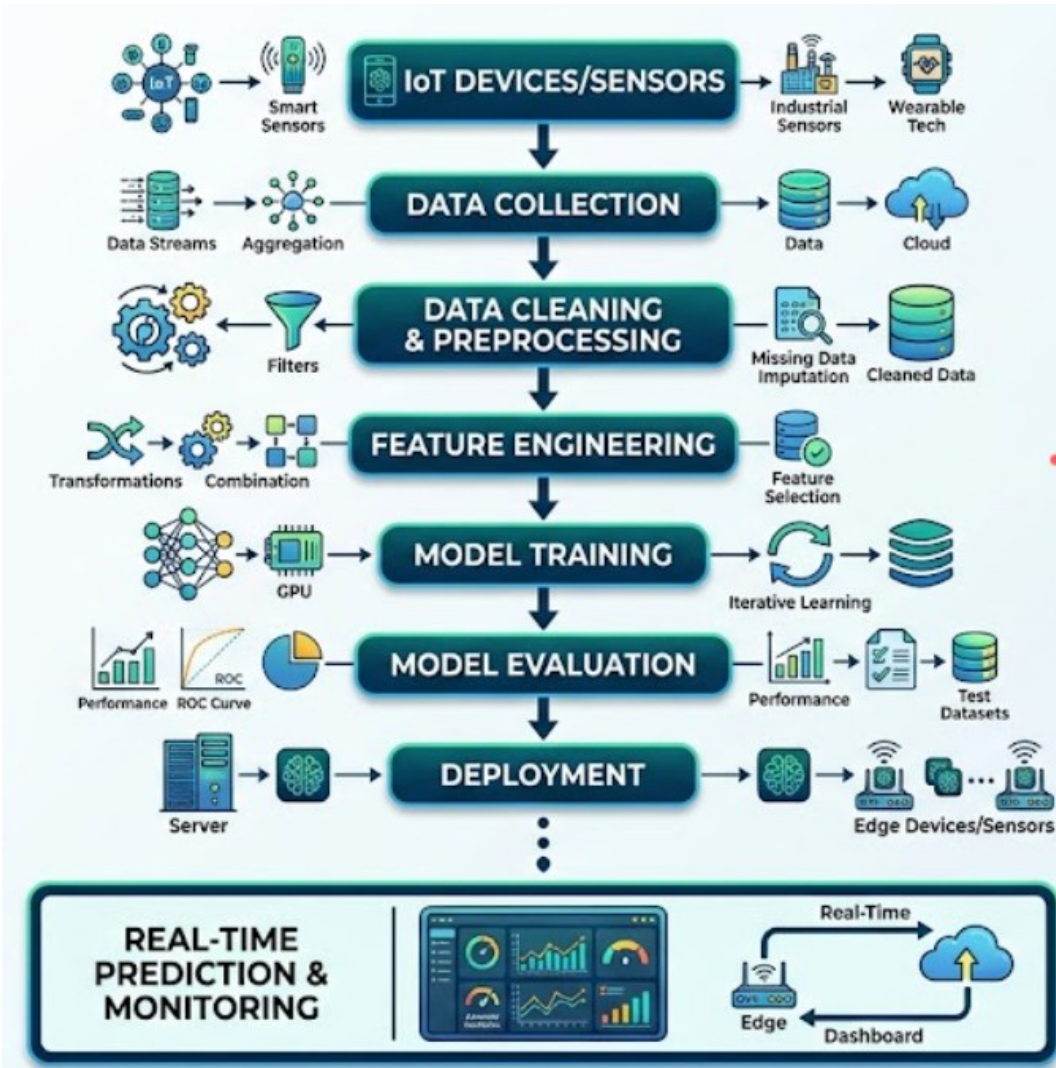
AI becomes operational

Predictions become actions

Connected systems
become intelligent systems

Key Takeaways

12



Machine Learning in AIoT is a Continuous Lifecycle

Learners Should Remember

- AI learns from data
- Data quality is critical
- Feature engineering improves predictions
- Models must be evaluated carefully
- AIoT combines real-time data with real-time intelligence

Final Thought

- AIoT transforms connected systems into intelligent autonomous systems